

Topic A Part 1 Review [106 marks]

1.

[Maximum mark: 18]

25M.2.SL.TZ2.1
- Option A — Databases**

Alpha Hospital is situated in a large city and has over 1000 staff. It stores its data in a relational database.

Some of the tables in this relational database contain data about the hospital staff, their roles, and the hospital departments.

Staff roles include doctor, nurse, pharmacist, radiologist, and support staff. Staff can only hold one role.

Departments include Accident and Emergency, Critical Care, Medical, General Surgery, Orthopaedics, and Ophthalmology. Staff can work in several departments.

The ROLE table, STAFF table, and DEPARTMENT table are shown in **Figure 1**.

Figure 1: Diagrammatic relationship for the ROLE table, STAFF table and DEPARTMENT table

ROLE	STAFF	DEPARTMENT
RoleID	StaffID	DepartmentID
PayGrade	RoleID	ManagerID
	DepartmentID	Building
	FirstName	
	Surname	
	DateOfBirth	

- (a.i) State the primary key in the STAFF table.
- [1]

Markscheme
<i>Award [1 max]</i> <i>StaffID;</i>

Examiners report
Both parts of this question were well-answered by almost all students.

(a.ii) State a foreign key in the STAFF table.

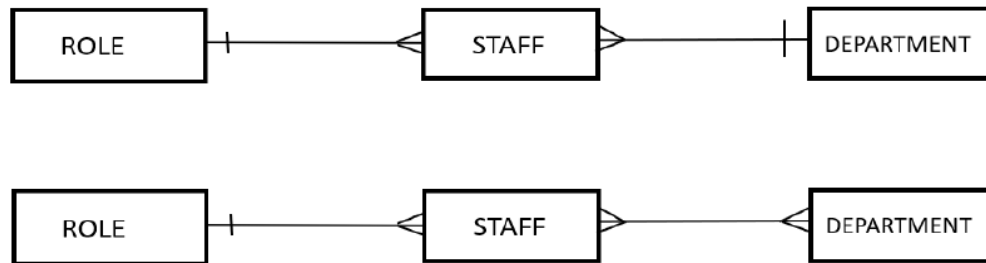
[1]

Markscheme
<i>Award [1 max]</i> <i>RoleID</i> <i>DepartmentID</i>

(b) Describe the relationships between the three tables in **Figure 1**.

[2]

Markscheme
<i>Award [2 max]</i> <i>Award [1] for the correct relationship between role and staff tables</i> <i>Award [1] for the correct relationship between department and staff</i> ROLE and STAFF: one to many STAFF and DEPARTMENT: one to many//accept many to many Note: if a diagram is seen ignore the symbols and award marks for the relationships only



Examiners report

The majority of students were able to correctly identify the relationship between STAFF and DEPARTMENT but that of ROLE and STAFF was incorrectly identified as *one-to-one* by a good number of students. Perhaps, the statement that *Staff can only hold one role* in the preamble was misinterpreted.

(c) Outline what a query is used for in a database.

[2]

Markscheme

Award [2 max]

The query provides a virtual representation of the database;
This shows the relationship between the different attributes;

A query searches/provides a filter view of the database;
based on the criteria set in the query;

A query helps to manipulate the data in the database;
Using INSERT/DELETE/UPDATE statement;

A query is used to aggregate/summarize data;
Using functions e.g. SUM(), AVG(), COUNT() etc.;
And enables grouping/sorting of data;

A query can alter the schema/structure of the database;
By modifying fields/data types of relations;
By creating/altering the relationship between tables/relations;

Note: *Do not award marks across different clusters*

Examiners report

The majority of students could identify the use of a query to retrieve data, but most of them could not make reference to the need for some filtering criteria. Only a handful of students referred to the other uses of a query like data manipulation, data aggregation, and schema creation/altering.

- (d) Identify the steps to create a query to list the staff with the surname Waters who are on pay grade 17. The query must display **only** FirstName, Surname, and PayGrade.

[4]

Markscheme

Award [4 max]

Award [1] for selecting FirstName, Surname and PayGrade

Award [1] for correct JOIN (or mentioning the joining of) the tables STAFF and ROLE)

Award [1] for the correct JOIN condition (STAFF.RoleID = ROLE.RoleID)

Award [1] for identifying both conditions (Surname and PayGrade condition)

Example 1

```
SELECT STAFF.Firstname, STAFF.Surname, Role.PayGrade
```

```
FROM STAFF INNER JOIN Roles ON STAFF.RoleID = ROLE. RoleID
```

```
WHERE STAFF.Surname = 'Waters' AND ROLE.PayGrade = 17;
```

Example 2

```
SELECT STAFF.Firstname, STAFF.Surname, ROLE.PayGrade  
  
FROM STAFF, ROLE  
  
WHERE STAFF.RoleID = ROLE.RoleID  
  
AND STAFF.Surname = 'Waters' AND ROLE.PayGrade = 17;
```

Note: *Accept equivalent in Structured English*

Examiners report

Most students were able to identify the fields and the required tables for the query, but they failed to describe/write an appropriate JOIN condition. The majority of responses included the correct filtering condition in the WHERE clause.

- (e) Outline why an integer is an appropriate data type for the PayGrade field.

[2]

Markscheme

Award [2 max]

The pay grade values are whole numbers;
The use of an integer reduces the size of the field;
Listing/sorting the staff in order of the PayGrade/increasing or decreasing of PayGrade;
Allows calculation and mathematical functions;

Examiners report

Many students were able point out the fact that the data in the field has whole numbers, but only a handful of them could proffer another reason.

- (f) Explain **two** ways in which the database administrator can ensure the privacy of the hospital's staff data.

[6]

Markscheme

Award [6 max]

Award [3 max] x [2]

Award [1] identifying the method

Award [1] for explaining how privacy is ensured using this method

Award [1] for application of the method to hospital staff data (accept other suitable examples)

By separating sensitive data (*paygrade and possibly age/DoB*) into a separate table;
Minimum number of people have access to sensitive data;
And only using the unique identifiers of the *staff* as a reference point;

The use of different levels of access/authorization;
Meaning the minimum number of people have access to this data;
For example, password protect/lock/restrict access to Role table;

Encrypting the stored data in the database (or specific columns in tables);
The data is scrambled/converted to cipher text (and cannot be understood without a key);
For example, only the employees with the key can decrypt the data;

Data anonymisation/masking/obfuscation;
This removes or transforms Personally Identifiable Information (PII) in the data;
For example, suppressing certain characters in the staff pay Grade etc;

Database views can be used to limit data access;
By exposing only specific columns or rows from a table;
For example a view can be created for a dept manager to see the data of staff in their dept only;

Examiners report

Majority of the students could identify the methods for ensuring data privacy, but ultimately struggled to explain how these methods ensure privacy and give linking examples to the hospital staff data.

2. [Maximum mark: 15] 25M.2.SL.TZ2.2
Database design is a complex process that takes place in a range of phases.
Different phases of database design use different schema.

- (a) Describe the difference between a conceptual schema and a logical schema.

[2]

Markscheme

Award [2 max]

Award [1 max] description of conceptual schema

Award [1 max] description of logical schema

The conceptual schema is a high level/least detailed representation of the database;

Conceptual is sometimes described as the “what” Model;

Involves identification of entities and (high level) relationships between them;

The logical schema is more detailed than a conceptual schema and shows details such as field names;

Logical schema is sometimes described as the “How” Schema;
Logical schema is developed from the Conceptual schema;

Examiners report

Many students could not show a clear difference between conceptual schema and logical schema. Only a handful of students showed understanding of the fact that both include entities and relationships.

- (b) Explain the importance of a data definition language in implementing a data model.

[2]

Markscheme

Award [2 max]

Award [1 max] explanation of why DDL is important

Award [1 max] example of use

A DDL is used to specify the schema of a database;

A DDL allows you to define the tables, fields, and set datatypes (e.g. CREATE TABLE);

A DDL allows you to specify keys/create relationships;

A DDL allows you to modify schema (e.g. DROP, ALTER);

Note: *Award 1 mark for any valid DDL command term*

Examiners report

Many students demonstrated little understanding of what a data definition language is and its uses in database creation.

- (c) Explain why data modelling is used during the development of a database.

[4]

Markscheme

Award [4 max]

Award [1] explanation and [1] for example x 2

It helps to identify the entities/tables in the database;
If incorrect, they will not support the purpose of the DB;

The attributes of the tables should be necessary and sufficient for the purpose;
To avoid inefficiencies in the operation of the database;

The keys for accessing the tables need to be identified;
To ensure the user can easily access the data;

The relationships between tables are identified;
To enable the user to perform complex queries across several tables;

Normalization during data modelling reduces data duplication;
Which reduces data anomalies;
And saves storage space;

Enables the database structure to be understood by other stakeholders/analyst/programmers;
Which makes database maintenance easier;

Examiners report

Students showed understanding of what data modelling entails, but most responses were either generic or vague: they failed to identify specific aspects of data modelling and their importance to the overall purpose and effective use of the database.

- (d) Explain why both data validation and data verification are required to ensure the correctness of the data within a database.

[3]

Markscheme

Award [3 max]

Data validation is an automated process that ensures input meets the data entry rules)/Data validation is an automated process that ensures input data is reasonable/valid;

Data verification is the checking of data to ensure it is the input intended; Using both techniques will provide the optimal solution, data validation can be applied to fields where the values can be easily determined (for example, no date of birth before a specified date), then data verification allows checking where data validation is not possible/appropriate (for example, a person's surname);

Award one mark for each definition [max 2] and one mark for linkage [1]

Examiners report

Most students showed understanding of data validation, but a good number of them inappropriately stated that data verification checks correctness of data. The linking mark was rarely earned.

- (e) Outline how data integrity is maintained during a database transaction.

[2]

Markscheme

Award [2 max]

Integrity is maintained by no changes being made until the transaction is complete;

If the transaction cannot be completed it is rolled back (to the original state);

Atomicity (all or nothing): change made or no changes made;

Consistency ensures a transaction does not put the database in an invalid state;

Rules and integrity constraints (validation, entity and referential integrity) are automatically enforced before a transaction is committed;

Preventing anomalies like orphan records and erroneous data;

Isolation prevents interference between concurrent transactions;

Which ensures intermediate/uncommitted values of one transaction are not visible to other transactions;

Through the use of locks and other concurrency control measures;

Preventing errors like the lost update, dirty reads, phantom reads etc;

Durability ensures that the effects of a committed transaction are permanent (irrespective of a subsequent system failure);

Transaction logs record all the changes made during transactions;

If a system failure occurs, the committed transactions in the log are reapplied to the database during recovery;

Data integrity is ensured by preventing data loss;

Note: Award 1 mark if a student only refers to ACID properties without any expansion

Examiners report

The majority of students generally referred to the use of ACID properties of a transaction to ensure data integrity. The expectation was to give an outline of how one of the properties ensures data integrity. Most of them did not pay attention to the command term.

- (f) Outline the role of relational integrity in maintaining data consistency within a database.

[2]

Markscheme

Award [2 max]

Referential integrity refers to relationships between tables in a database;
Referential integrity is maintained by the connection between the primary key in one table and the foreign key in another;
Which ensures records are appropriately updated (e.g. FK value cannot be changed to a value that does not exist as a PK);
It cascades updates and deletes from the PK table to the FK table;
To prevent update anomalies and orphan records;

Examiners report

This was well-answered by a large number of students.

3. [Maximum mark: 12]

25M.2.SL.TZ2.3

Alpha Hospital uses a database application to book appointments. Information can be shown in different formats.

Figure 2 shows an example of a patient's appointments displayed in the application.

Figure 2: A patient's appointments

< Back Patient appointments + ↺

Patient number

First name

Preferred name

Surname

Date of birth

Age

Form List

Select the appointment for details and directions to the building and room.

24 Jan 2024	09:30	Dr L. Ross	General Surgery
26 Jan 2024	10:00	Dr N. Friend	Radiology
28 Jan 2024	09:30	Dr L. Ross	General Surgery

👤 📍 📁 📅 ↩

The Age field is a derived field.

- (a.i) Outline **one** reason why a derived field would be used in a database.

[2]

Markscheme

Award [2 max]

A derived field is calculated by using data that exists within the database;
That means the data does not have to be input/does not take up storage space;

A derived field has currency as it is calculated/derived at the time so accounts for changes in date etc;

Examiners report

This was satisfactorily answered as most students identified saving storage space with the use of derived fields. A handful of more able students presented complete responses.

(a.ii) Describe how the derived field Age would be calculated.

[2]

Markscheme

Award [2 max]

Award [1] for identifying the data from the database;

Award [1] for the correct formula to create the derived data;

[DateOfBirth] = SELECT DateOfBirth FROM Table

Age = YEAR([Date]) – YEAR([DateOfBirth])

SELECT DATEDIFF(YYYY, DateOfBirth, GetDate()) AS Age FROM Table

SELECT TIMESTAMPDIFF(YEAR, DateOfBirth, CURDATE()) AS Age FROM Table;

SELECT DATE_FORMAT(NOW(), '%Y') - DATE_FORMAT(DOB, '%Y') AS age FROM Table

Accept similar structure

Accept answers in structured English example:

Select Date of Birth from the patient table;

Subtract the year of the birth date from the current year to get the "Age";

Examiners report

The majority of the students could identify the use of the current date and the date of birth field in the database to calculate the age, but did not realise that the difference between two dates gives another date (and not an integer value that represents the age).

Patient information can be represented in the following format:

PATIENT (PatientID, FirstName, Surname, PreferredName, DateOfBirth)

- (b) Outline the difference between first normal form (1NF) and second normal form (2NF).

[2]

Markscheme

Award [2 max]

The prerequisite for 2NF is 1NF;
1NF has no prerequisite;

The focus of 1NF is to eliminate repeating groups, ensure atomicity of data values and ensure entity integrity;
The focus of 2NF is to ensure full functional dependency (remove partial dependencies);

1NF applies to any data table;
2NF only applies to tables with composite keys;

Examiners report

This was well-answered by the majority of students.

- (c) Construct a database in third normal form (3NF) for all of the data shown in **Figure 2**. You should use database notation as shown in the PATIENT table.

[6]

Markscheme

Award [6 max]

Award [1] for identifying three tables with correct fields (without PK and FK)

Award [1] for the DEPARTMENT table (correct fields and PK identified)

Award [1] for the DOCTOR table (correct fields and PK identified, without FK)

*Award [1] for including FK (DepartmentID) in DOCTOR/APPOINTMENT table **but not both***

Award [1] for identifying the PK in the APPOINTMENT table

Award [1] for including FK (PatientID, DoctorID) in APPOINTMENT table

Award [1] for all fields correct in APPOINTMENT table

Example 1

PATIENT (PatientID, FirstName, Surname, PreferredName, DateOfBirth)

DEPARTMENT (DepartmentID, departmentName)

DOCTOR (DoctorID, FirstName, Surname, DepartmentID*)//accept
DoctorName

APPOINTMENT (PatientID*, DoctorID*, date, time)

Example 2

PATIENT (PatientID, FirstName, Surname, PreferredName, DateOfBirth)

DEPARTMENT (DepartmentID, departmentName)

DOCTOR (DoctorID, FirstName, Surname)//accept DoctorName

APPOINTMENT (PatientID*, DoctorID*, date, time, DepartmentID*)

Award the mark for FK if not explicitly indicated as long as the name matches the PK

Accept an additional field as PK in APPOINTMENT:

APPOINTMENT (AppointmentID, PatientID*, DoctorID*, date, time)

*Accept a **dateTime** field in APPOINTMENT instead of **date** and **time** separately.*

Accept LOCATION in place of DEPARTMENT (and LocationID in place of DepartmentID):

LOCATION (LocationID, LocationName)

Note: *Apply the MS according to example 1 or example 2*

Note: *The relation names do not matter; check the fields used*

Examiners report

Most of the students showed understanding of the basic concepts of normalization and were able to identify the required tables. Others inappropriately included tables from the previous preamble, and this complicated the process for them. A good number of them lost marks because they did not identify the primary and foreign keys for the relations.

4. [Maximum mark: 8]

24N.2.SL.TZ0.1

After a crime has been committed, police officers carry out a number of interviews.

In each interview, a police officer will interview a witness (a person who saw the crime being committed).

- A witness may be interviewed many times.
- A police officer may carry out many interviews.

- (a) Construct an entity-relationship diagram (ERD) that shows the relationships between the interview, the police officer and the witness.

[2]

Markscheme

Award [2 max]

Award [1] for 3 correct tables or 2 correct tables with a 1-M

Award [2] for three correct tables and relationships



Examiners report

This was well-answered by a large majority of the students.

When creating a relational database, such as a police database, it is important to consider referential integrity.

- (b) Identify **three** reasons why it is important to enforce referential integrity.

[3]

Markscheme

Award [3 max]

Improves data accuracy / reduce record anomalies / improves data retrieval accuracy;

Improves data consistency / Maintain relationships / Ensure records in a

linked table relates to data in another table (via primary key and foreign) to allow extraction of data from different table;

Ensure that a deleted record does not leave hanging records unlinked to other tables;

To simplify update / maintenance through cascade delete / update;

Note: Accept related example such as if a police officer who interviewed a witness, is deleted, they cannot be identified from the interview record.

Examiners report

It was easy for student who understand the concept to obtain maximum marks but a good number of them did not get any mark all. Most students could identify improvement in data consistency as a reason for ensuring referential integrity.

(c) Explain how a query can provide a view of the database.

[3]

Markscheme

Award [3 max]

A name is assigned to a new view/CREATE VIEW view_name AS;
Fields (columns) are selected/SELECT field1, field2, FROM table(s);
The condition is specified/WHERE condition/filter applied;
Which will reduce the subset of data to include required data only;

A query/filter provides a subset of the database;
Which can be saved and used regularly (e.g. weekly updates);
The view is not part of the physical schema;
The view can be organised/sorted according to the reporting needs;

Note: Do not award marks from different clusters.

Examiners report

Majority of the students demonstrated the understanding that a query presents a view as a subset of the database containing the required data. More able students were able to outline the need for projection and selection as part of query construction.

5. [Maximum mark: 15]

24N.2.SL.TZ0.2

(a) Define the term *secondary key*.

[1]

Markscheme

Award [1 max]

It is a non-identifying field used to find row in a table;
It is an additional key which can be used to locate specific data;
It is a candidate key that is not the primary key;

Examiners report

Most students were able to define the term secondary key. A common error was equating it to foreign key.

In the police database, a WITNESS table, CRIME table and OFFICER table have been created, but are not shown. The primary keys for these tables are WitnessID, CrimeID and OfficerID respectively.

When a police officer interviews a witness, they ask for a statement about the crime.

Sometimes, a witness may need to be interviewed more than once by the same police officer.

A record of each interview is placed in the INTERVIEW table:

INTERVIEW(WitnessID, CrimeID, OfficerID, Statement, CrimeLocation, DateTimeStatement, PoliceStation, StationPhone)

The foreign keys WitnessID, CrimeID and OfficerID in the INTERVIEW table form a composite primary key for this table.

(b.i) Outline why the use of WitnessID, CrimeID and OfficerID is not a suitable composite primary key.

[2]

Markscheme

Award [2 max]

The three combined keys do not uniquely identify each interview;
A witness can be interviewed more than once by the same officer for the same crime;
A primary key (e.g. INTERVIEWID) is more suitable than a composite key with many fields / A composite key would need to include DateTimeStatement;
Indexing works better with one key than a composite key with multiple fields;

Examiners report

Majority of the students determined that the composite key could not uniquely identify a record due to duplication of the attributes. Above

average students were able to give the bases for the repetition and outline the efficiency of a single-attribute primary key in record retrieval.

- (b.ii) Outline why the INTERVIEW table is not in 3rd Normal Form (3NF).

[4]

Markscheme

Award [4 max]

The table is not in 1NF/2NF so cannot be in the 3NF;
There is no (unique) Primary Key;
There is partial dependency // non-key attribute depends on part of primary key only...
... Example: CrimeLocation is dependent on CrimeID only (and not the composite key);
The table has transitive/non-key dependency; ...
... Example: StationPhone is dependent on Station;

Examiners report

There were some excellent responses with students showing good understanding of partial dependency and transitive dependency. However, some students mixed up the supporting examples for the two dependencies. A good number of students did show any understanding of the requirements for 3NF.

Figure 1 includes data from two crimes.

Figure 1

Fields	Crime 1	Crime 2
CrimeID	56326	56543
CrimeType	Theft	Computer misuse
OfficerID and OfficerName	OfficerID OfficerName 0071 Delphine Allais 0056 Eric Demoncheaux	OfficerID OfficerName 0099 Christelle Chades 0082 Arthur Katalayi
CrimeAddress	18 Rue St Michel, 75011 Paris	127 Rue St Jacques, 75011 Paris
CrimeDateTime	2022-06-21 03:33:00	2022-06-21 02:27:00
EvidenceType	Statement Statement Photograph	Statement Statement Video
EvidenceDesc	Scott Ngatai statement Zixin Yang statement Photograph of broken window	Ciara Alinec statement Zixin Yang statement CCTV video 02:20 to 03:00
VictimID (VID) VictimName, and VictimPhone	VID VictimName VictimPhone HK20 Hana Kim 01029238484 ZY33 Zixin Yang 01029968565 SN21 Scott Ngatai 01029451869	VID VictimName VictimPhone CA19 Ciara Alinec 01021103431 ZY33 Zixin Yang 01029968565 HK20 Hana Kim 01029238484

The following database notation represents the fields in **Figure 1**.

CRIME(CrimeID, CrimeType, OfficerID, OfficerName, CrimeAddress, CrimeDateTime, EvidenceType, EvidenceDesc, VictimID, VictimName, VictimPhone)

The data type for VictimPhone is alphanumeric.

- (c) Outline why an alphanumeric data type is used for the VictimPhone field.

[2]

Markscheme

Award [2 max]

Phone number is not a value/amount/magnitude;
 May start with 0 and get removed if integer type is used;
 May include brackets and/or + symbol;
 May need to search for part of the number;

Examiners report

This was a challenging question for majority of students with only a handful of them obtaining both marks.

- (d) Construct the database in 3rd Normal Form (3NF) for all of the data shown in **Figure 1**. You should use database notation as shown in the CRIME table.

[6]

Markscheme

Award [6 max]

Award 1 mark for identifying 4 tables with correct fields without PK;

Award 1 mark for EVIDENCE with EvidenceID as PK and correct fields;

Award 1 mark for CRIME with CrimeID as PK and correct fields;

Award 1 mark for VICTIM with VictimID as PK and correct fields;

Award 1 mark for OFFICER with OfficerID as PK and correct fields;

Award 1 mark for identifying CrimeID as FK in EVIDENCE;

Award 1 mark for CRIMEOFFICER with correct fields and PK;

Award 1 mark for VICTIMCRIME with correct links and PK;

CRIME(CrimeID, CrimeType, CrimeAddress, CrimeDate, CrimeTime)

CRIMEOFFICER(CrimeID, OfficerID)

EVIDENCE(EvidenceID, EvidenceType, CrimeID*, EvidenceDesc)

VICTIMCRIME(CrimeID*, VictimID*)

VICTIM (VictimID, VictimName, VictimPhone)

OFFICER (OfficerID, OfficerName)

Accept CrimeDateTime in CRIME

Accept a primary key for VICTIMCRIME

*Accept EvidenceTypeID in EVIDENCE as a foreign key
Students may split OfficerName to OfficerFirstname and OfficerSurname*

Examiners report

Most students could easily identify the base relations together with correct primary keys (PK) but the identification of the bridge tables was not straightforward for most students. A good number of students lost marks for not identifying the PK for each relation.

6. [Maximum mark: 22] 24N.2.SL.TZ0.3
Two police officers attempt to update a witness statement at the same time.

- (a) Describe how data locking can be used to deal with this situation.

[2]

Markscheme

Award [2 max]

Data locked could be applied to a record/row as soon as it's opened in edit mode/when a transaction is being carried out;
A second officer would have read-only access/would not be given editing rights;
Until the edit is complete/until a timeout period has been applied;
Locks can be applied and removed in two phases to ensure serial sequences (two phase locking);

Examiners report

Majority of the students showed a general understanding of the procedure required in updating the witness statement, but the concept of row/record locking was missing in most responses.

Many countries have a centralized crime database.

- (b) Discuss the advantages and disadvantages of using a centralized crime database.

[5]

Markscheme

Award [5 max]

Advantages

It eliminates data duplicates / update anomalies;

And improves the integrity / consistency of data;

Improves quick access / sharing of data (between different law enforcement agencies);

As there is no need to look at different sources;

Thus, reducing time required to solve crimes;

Crimes committed in other regions can be cross-referenced;

To improve the accuracy of investigations;

It increases available (historical) data;

And this improves the insight from data analytics;

Disadvantages

One central point can be attacked by hackers;

Leading to complete data loss / privacy issues;

Central computer is a single point of failure;
Which can cause system downtime (hampering crime investigation);

Increased bureaucracy (as getting access to the data might require several levels of security clearances);
Can prevent timely access to crime data;

Data can be intercepted (and replayed) during transmission;
Compromising the investigation;
Raising privacy issues leading to data protection breaches;

Costs of development / maintenance can be high;
In terms of the hardware / security / transmission infrastructure;

Conclusion
A valid conclusion with a justified opinion.

Marks [2] + [2] + [1]

Examiners report

Most students could provide the advantages and disadvantages of centralized databases, but many others incorrectly assumed a centralized database is the same as a data warehouse. Majority of the responses have no conclusion as required by the command term.

Data mining and data matching algorithms are used to analyse crime data.

- (c.i) Explain why data mining might be used to extract information from crime data held in a police database.

[4]

Markscheme

Award [4 max]

Explanation of data mining (the use of pattern recognition logic to identify trends within a large dataset);

DM finds hidden patterns and correlations within crime data;

DM analyses historical crime data which reveals trends;

DM enables the exploration of relationships and associations between different entities/uses link analysis;

DM can build predictive models (based on historical data);

DM can identify anomalies or outliers in crime data;

For example, the algorithm might predict the location/time of the next crime from a series of related crimes.

Note: only award full marks if there is a valid reference to the scenario. Accept other examples. Award 3 marks max if there is no reference to the scenario.

Examiners report

Majority of the students showed understanding of the use of data mining and provided valid examples; however, they could not provide further details relating to specific strategies that can be applied to the crime data.

- (c.ii) Explain, using an example, how data matching could be applied to the crime data.

[4]

Markscheme

Award [4 max]

Explanation of data matching (the comparison of different sets of data with the aim of identifying links/similarity between them);

Data matching identifies records that correspond to the same entities (e.g.

suspects, location, type of crime);

Data matching can be run across multiple databases such as hospital records, insurance claims, vehicle registration;

Unrelated crimes or entities may be grouped together (classified) by common data;

The results of data matching can be further used to reveal connections between past and present crimes;

Examples may include:

CCTV footage may link a car to more than one investigation;

Suspect identification – matching physical characteristics to several crimes;

Alias detection – criminals using multiple aliases to hide their identity;

Network analysis – linking criminals together into a crime syndicate;

Note: only award full marks if there is a valid reference to the scenario. Accept other examples. Award 3 marks max if there is no reference to the scenario.

Examiners report

This was quite a challenging question for most students. While some confused it with data mining, others likened it to a regular database query that retrieves tuples with matching criteria.

Legislation has been created to protect the rights of individuals whose data is stored electronically. Most countries have similar regulations to the General Data Protection Regulation (GDPR) found in Europe.

One requirement of legislation is that data must be secure.

(d.i) Outline **two** reasons why a password-only approach would not be secure.

[4]

Markscheme

Award [4 max]

Dictionary / brute force attack;

Because of weak passwords e.g., the user only used letters / numbers;

Users choose the same password for everything;

Leaked password from hacked software / written down / obvious password;

Keylogging may be used to obtain password;

if two or three-factor authentication is not used / an on-screen keyboard isn't used;

A password is susceptible to a MITM attack;

If no encryption is used;

Marks [2] + [2]

Examiners report

Generally, this question was not well-answered: most responses were vague. Most students identified the use of brute force but failed to state that this is only effective with weak passwords. More able students were able to outline the need for multiple factor authentication (MFA).

- (d.ii) Identify **three** ways that legislation, such as GDPR, protects individuals whose data is stored electronically.

[3]

Markscheme

Award [3 max]

By providing a framework for ethical storage of data/requiring companies to fairly and lawfully process data;
By requiring companies to register with the data protection governing body;
By enforcing (financial) penalties on organisations that fail to follow guidelines;
Holding data for the period of time that is required only;
Using data collected for intended purpose only;
Not sharing data without permission (outside organisations/counties);
Giving individuals the right to view data about themselves;
Storing data that is adequate, relevant and not excessive;
Ensuring data is accurate and up-to-date;

Examiners report

Majority of the students were unable to identify ways in which legislation, such as GDPR, protects individuals whose data is stored electronically. Most of them produced off course responses that referred to what organisations must do to protect personal data.

7. [Maximum mark: 16]

24M.2.SL.TZ0.1

Environmental systems and societies (ESS) students are collecting data about the plant species found on sand dunes as part of their internal assessment. The data is collected from 10 sites using a paper form (**Figure 1**).

The form shown in **Figure 1** is used to input data into the **Environment** database.

Figure 1: An example of a data collection form.

FIELD STUDY CENTRE

[LOGO]

Survey date: 14/10/2019

Sample site: 9

 Aspect: South

 Gradient: 10

Established dune, south facing

Comments: None

Species	Present	Coverage %
Marram grass		0
Sand couch grass		0
Sea holly		0
Sea spurge		0
Sea buckthorn		0
Gorse	✓	40
Blackthorn	✓	40

(a) State the data type for:

(a.i) Species

[1]

Markscheme

Award [1 max].

String or Text;

(Accept also Varchar, Alphanumeric)

Examiners report

This was well-answered by a large majority of students.

(a.ii) Gradient

[1]

Markscheme

Award [1 max].

Real;

(Accept also Integer, Float, Number, Numeric)

Examiners report

This was also well-answered. A common error was stating programming-related data types.

(b) Outline **one** way that data validation could be carried out on the gradient attribute.

[2]

Markscheme

Award [2 max].

Range check;

... to ensure input is between -90 to $+90$;

Presence check;

... to ensure a value has been input;

Type check;

... to ensure the input is numeric;

Data validation is checking to see if the data entered is within an appropriate range;

For example, to check that gradient is between -90 to $+90$;

Note: The question asks for **one** way. Read whole response and mark to student's advantage

Examiners report

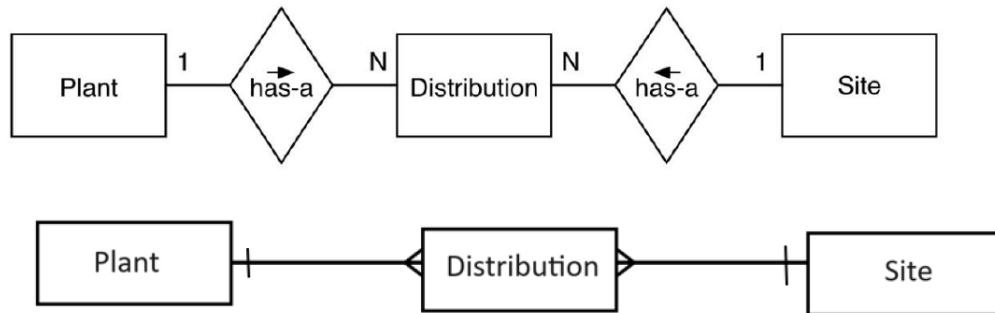
Many students were able to identify an appropriate validation check for the gradient attribute, but giving an outline of the stated validation check in the context was challenging. Some responses included verification checks instead.

- (c) Construct an entity relationship diagram (ERD) for the Plant, Site and Distribution tables.

[3]

Markscheme

Award [3 max].



Award [1] for three entities (tables - in any order);

Award [1] for Plant to Distribution (1 to many);

Award [1] for Site to Distribution (1 to many).

Note: Allow alternative ERD layouts

Examiners report

The majority of the students were able to correctly represent the entities in the scenario described. However, representing the relationship between the entities was a challenge. A few students identified one-to-many relationships but incorrectly indicated the prime and the dependent entities on the diagram. A few other students struggled with the use of appropriate symbols, with some incorrectly using the diamond symbol to represent entities.

- (d) Outline why a composite primary key is used for the Distribution table.

[2]

Markscheme

Award [2 max].

The Plant_ID and Site_ID can be repeated / not unique // Many plants can be in a site and many sites can have the same plants // No one attribute in Distribution table can uniquely identify the record;
Use of multiple fields uniquely identifies a tuple/record;
And removes the need to add a new ID field as Primary key (which would waste storage space);

Examiners report

Many students demonstrated the understanding that a composite key involves two or more fields but the idea of uniqueness for a primary key needed to be included in their responses: most of them incorrectly stated that Plant_ID and Site_ID together form a valid composite key for the relation. It is obvious these students need to learn how prime attributes are shown in relations since the primary key has been clearly indicated in the table.

- (e) Identify the steps to create a query to calculate the total number of sites where gorse has been found from the samples carried out on 14 October 2019.

[4]

Markscheme

Award [4 max].

Award [1] for all relevant tables used (Plant and Distribution);

Award [1] for use of count on SiteID;

Award [1] for correct species and date conditions;

Award [1] for correct condition for connecting the two tables.

```
SELECT COUNT(Site_ID) FROM Plant, Distribution WHERE Species =  
"Gorse"  
AND DATE = '14/10/2019' AND Plant.Plant_ID = Distribution.Plant_ID
```

or

//Get the Plant_ID from the Plant table:

```
@PlantID = (SELECT Plant_ID FROM Plant WHERE Species = 'Gorse')
```

// @PlantID = 6 at this point; They can use 6 here instead of @PlantID

```
SELECT COUNT(Site_ID) FROM Distribution WHERE Plant_ID = @PlantID  
AND DATE="14/10/2019"
```

or

```
SELECT COUNT(Distribution.Site_ID)  
FROM Plant INNER JOIN Distribution  
ON Plant.Plant_ID = Distribution.Plant_ID  
WHERE Plant.Species = "Gorse" AND Distribution.Date = '14/10/2019'
```

or

(Accept also a description, e.g.)

Create a JOIN between the Plant and Distribution tables based on the Plant_ID field and select records with Species = 'Gorse' and Date = '14/10/20219'. Count the number of records in the table.

Note: Accept other date formats. Accept double or single quotes for strings.

Examiners report

This was the most challenging question for the students. Only a handful of students used the COUNT aggregate function. Most of them were able to identify the tables required for the query but they failed to describe/write an appropriate JOIN condition. Majority of the responses did not include the filtering condition on the Species field; they went on to incorrectly use the Plant_ID corresponding to the given species, and consequently lost the WHERE CLAUSE mark.

- (f) Explain how data consistency can be maintained in the **Environment** database.

[3]

Markscheme

Award [3 max].

Any data written to the environment database must be valid according to all defined rules; Referential integrity;

... all foreign keys should have a primary key;

Cascades;

... changing data in one table should be carried through to related tables //

Update of the data must happen uniformly across all occurrences;

Normalise the database

... to remove/reduce redundant data // to remove repeated data;

Triggers;

... to automatically invoke updates to tables as required;

... all data must be sensible/conform to set rules before being stored;

Validation checks;

Verification checks;

Data updates use transactions to ensure automatic rollback on failure;

Examiners report

The majority of students responded appropriately to the question. Many of them wrote copiously about data consistency without being particular about the steps that must be taken to ensure it.